

Conditional Branching: if, '?' - Simple Hinglish Summary

Source: <https://javascript.info/ifelse>

Main baat

Kabhi kabhi **alag conditions ke hisaab se alag kaam** karna padta hai. Iske liye `if` statement aur conditional operator `?` (question mark / ternary) use hote hain.

1. `if` statement

`if(...)` bracket ke andar ki **condition check** karta hai. Agar result `true` ho to code block chalta hai.

javascript

```
let year = prompt('In which year was ECMAScript-2015 specification published?');  
  
if (year == 2015) alert( 'You are right!' );
```

Ek se zyada statements chalane ho to **curly braces** `{ }` lagane padte hain:

javascript

```
if (year == 2015) {  
    alert( "That's correct!" );  
    alert( "You're so smart!" );  
}
```

Recommendation: Ek hi statement ho tab bhi **hamesha** `{ }` lagao. Readability badhti hai.

2. Boolean conversion

`if (...)` condition ko **boolean me convert** karta hai.

- Falsy values** (`false` ban jati hain): `0`, `""`, `null`, `undefined`, `NaN`
- Truthy values** (`true` ban jati hain): baaki sab

javascript

```
if (0) {  
    // kabhi nahi chalega (0 falsy hai)  
}  
  
if (1) {  
    // hamesha chalega (1 truthy hai)  
}
```

Pehle se calculate ki hui boolean value bhi de sakte ho:

```
javascript  
  
let cond = (year == 2015);  
  
if (cond) {  
    ...  
}
```

3. `else` clause

Condition **falsy** ho to `else` block chalta hai. (Optional hai.)

```
javascript  
  
if (year == 2015) {  
    alert( 'You guessed it right!' );  
} else {  
    alert( 'How can you be so wrong?' );  
}
```

4. `else if` (kai conditions)

Kai variants check karne ho to `else if` use karo:

```
javascript  
  
if (year < 2015) {  
    alert( 'Too early...' );  
} else if (year > 2015) {  
    alert( 'Too late' );  
} else {  
    alert( 'Exactly!' );  
}
```

Pehle `year < 2015` check hota hai. Falsy ho to agli condition, aur sab falsy ho to aakhri `else`. Jitne chahe `else if` laga sakte ho. Aakhri `else` optional hai.

5. Conditional operator `(?)` (ternary)

Kabhi kabhi condition ke hisaab se **variable me value assign** karni hoti hai.

`if..else` se:

javascript

```
let accessAllowed;
let age = prompt('How old are you?', '');

if (age > 18) {
  accessAllowed = true;
} else {
  accessAllowed = false;
}
```

`(?)` se (chhota tarika):

javascript

```
let result = condition ? value1 : value2;
```

Condition truthy ho to `value1`, warna `value2` milta hai.

javascript

```
let accessAllowed = (age > 18) ? true : false;
```

- Ise "**ternary**" bhi kehte hain kyunki iske **3 operands** hote hain. JS me ye akela aisa operator hai.
- `age > 18` ke around brackets **zaruri nahi**, lekin **readability ke liye recommended** hain.

Note: Upar wale example me `(?)` ki zarurat hi nahi, kyunki comparison khud `true/false` deta hai:

javascript

```
let accessAllowed = age > 18;
```

6. Multiple (?)

Kai (?) ek ke baad ek laga kar **kai conditions** handle kar sakte ho:

javascript

```
let age = prompt('age?', 18);

let message = (age < 3)    ? 'Hi, baby!' :
               (age < 18)  ? 'Hello!'   :
               (age < 100) ? 'Greetings!' :
               'What an unusual age!';

alert( message );
```

Kaise kaam karta hai:

1. Pehle (age < 3) check hota hai. True ho to 'Hi, baby!'.
2. Nahi to (:) ke baad wali condition (age < 18), true ho to 'Hello!'.
3. Nahi to (age < 100), true ho to 'Greetings!'.
4. Nahi to aakhri value 'What an unusual age!'.

Ye (if...else if...else) jaisa hi hai.

7. (?) ka galat use (avoid karo)

Kuch log (?) ko (if) ki jagah use karte hain:

javascript

```
(company == 'Netscape') ?
  alert('Right!') : alert('Wrong.');
```

Ye recommend nahi hai. Chhota zaroor hai, lekin **kam readable** hai. Aankhein code ko **vertically scan** karti hain, isliye multi-line blocks samajhna aasan hota hai.

Behtar tarika:

javascript

```
if (company == 'Netscape') {
  alert('Right!');
} else {
  alert('Wrong.');
```

```
}
```

Rule: (?) ka kaam ek value ya dusri value return karna hai, uska use usi ke liye karo. Alag code branches chalane ho to (if) use karo.

Practice Tasks (Answers)

1. (if ("0")) me alert dikhega? Haan. ("0") khali string nahi hai, isliye truthy hai.

```
javascript

if ("0") {
  alert( 'Hello' );    // chalega
}
```

2. JavaScript ka official naam:

```
javascript

let value = prompt('What is the "official" name of JavaScript?', '');

if (value == 'ECMAScript') {
  alert('Right!');
} else {
  alert("You don't know? ECMAScript!");
}
```

3. Sign dikhao (1, -1 ya 0):

```
javascript

let value = prompt('Type a number', 0);

if (value > 0) {
  alert( 1 );
} else if (value < 0) {
  alert( -1 );
} else {
  alert( 0 );
}
```

4. (if) ko (?) me badlo:

```
javascript

let result = (a + b < 4) ? 'Below' : 'Over';
```

5. `if..else` ko multiple `?` me badlo:

javascript

```
let message = (login == 'Employee') ? 'Hello' :  
  (login == 'Director') ? 'Greetings' :  
  (login == '') ? 'No login' :  
  '';
```

Quick Summary

Construct	Kab use karen
<code>if</code>	Condition true ho to code chalana
<code>else</code>	Condition false ho to alag code chalana
<code>else if</code>	Kai conditions check karni ho
<code>? :</code>	Condition ke hisaab se value assign karni ho

Yaad rakho:

- `{}` hamesha lagao.
- `0`, `""`, `null`, `undefined`, `NaN` falsy hain, baaki truthy (`"0"` bhi truthy hai).
- `?` sirf value return karne ke liye, code branches ke liye `if` use karo.